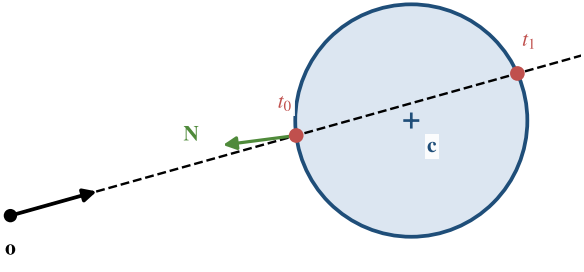
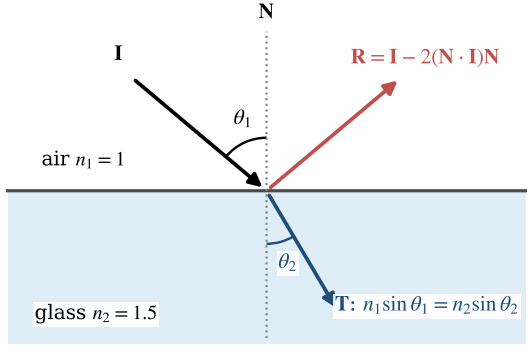
ray–sphere: nearest root t_0 is the hit



$$\mathbf{p}(t) = \mathbf{o} + t\mathbf{d}, \quad |\mathbf{p} - \mathbf{c}| = r$$

$$t = -b \pm \sqrt{b^2 - c}, \quad b = \mathbf{d} \cdot (\mathbf{o} - \mathbf{c}),\ c = |\mathbf{o} - \mathbf{c}|^2 - r^2$$

secondary rays: reflection and refraction



total internal reflection (glass to air) above $\theta_c = \arcsin(1/1.5) = 41.8^\circ$